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Diagnostic Accuracy of Dermoscopic Features in Acral Lentiginous Melanoma
A Systematic Review and Meta-Analysis

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IMPORTANCE Acral lentiginous melanoma (ALM) is a rare but aggressive melanoma subtype. Distinguishing ALM from acral nevi remains difficult, underscoring the need for reliable dermoscopic criteria.

OBJECTIVE To systematically evaluate dermoscopic patterns that differentiate ALM from benign acral nevi.

DATA SOURCES Embase, PubMed, Web of Science, MEDLINE, and the Cochrane Library were searched from January 1970 through March 2023 using controlled vocabulary (MeSH and Emtree) and natural language terms associated with acral melanoma, nevi, and diagnostic techniques. Searches were limited to English-language and human studies. Data analysis was conducted on November 1, 2024.

DATA EXTRACTION AND SYNTHESIS Following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, reviewers independently extracted study data. Pooled proportions and prevalence differences were assessed using random-effects models.

MAIN OUTCOMES AND MEASURES The main outcome was the prevalence of dermoscopic features, including parallel ridge, parallel furrow, latticelike, multicomponent, and other pigmentation patterns, with pooled differences between ALM and nevi.

RESULTS Forty-one studies were included, comprising 8845 acral nevi (from 35 studies) and 801 ALMs (from 18 studies). The parallel ridge pattern was the most sensitive and specific feature for ALM (79.6% [95% CI, 71.4%-87.8%] vs 0.6% [95% CI, 0.3%-0.9%] in nevi; pooled difference, −77.3%; 95% CI, −85.0% to −69.6%; $P < .001$). Multicomponent structures were also significantly associated with ALM (45.8% [95% CI, 19.3%-72.2%] vs 5.0% [95% CI, 3.7%-6.2%] in nevi; pooled difference, −38.5%; 95% CI, −65.7% to −11.2%; $P = .01$). Conversely, parallel furrow (51.8% [95% CI, 46.7%-56.8%] in nevi vs 8.9% [95% CI, 5.3%-12.5%] in ALM; pooled difference, 41.3%; 95% CI, 33.1%-49.5%; $P < .001$) and latticelike patterns (13.0% [95% CI, 10.6%-15.5%] in nevi vs 2.7% [95% CI, 1.0%-4.5%] in ALM; pooled difference, 8.9%; 95% CI, 4.2%-13.6%; $P < .001$) were significantly more common in nevi. Other features, including fibrillar, homogeneous, globular, reticular, other, and nontypical patterns, showed inconsistent or nonsignificant associations.

CONCLUSIONS AND RELEVANCE While the parallel ridge and parallel furrow patterns remain key diagnostic indicators of ALM and benign acral lesions, this systematic review and meta-analysis establishes the additional diagnostic significance of multicomponent and latticelike features. Nearly 20% of ALM lacked the parallel ridge pattern, demonstrating the need for new specific features for melanoma diagnosis. These findings provide evidence-based benchmarks to potentially improve diagnostic accuracy, guide dermoscopy training, and enhance early detection of ALM.

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Supplemental content

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Melanoma, a highly aggressive cutaneous cancer of melanocytes, accounts for only 1% of skin cancer diagnoses but is responsible for most skin cancer-related deaths due to its high metastatic potential.[1] The global health burden of melanoma continues to rise, with prevalence increasing by approximately 258% between 1990 and 2018.[2]

Acral lentiginous melanoma (ALM), which represents 2% to 3% of melanoma cases, is the fourth most common subtype of malignant melanoma. Acral sites are defined as those associated with the palms, soles, or nail unit. While the incidence of ALM in the US has remained relatively stable during the past several decades, it has been described as the most frequent melanoma subtype among patients with skin of color, including those of African, Asian, and Hispanic/Latino descent.[3] Prognosis is especially poor in advanced-stage disease, making early detection crucial for improving outcomes.[4-6]

Accurate recognition of ALM requires reliable clinical descriptors and dermoscopic criteria that distinguish malignant lesions from benign acral nevi, the most common pigmented lesions arising on acral sites.[7] Diagnosis typically involves a combination of clinical and dermoscopic examination, which is followed by histopathologic confirmation if warranted. Dermoscopy enhances the accuracy of direct clinical examination by revealing microscopic pigmentation patterns on the skin surface, several of which are diagnostically useful for melanoma. Prior studies have identified the parallel ridge pattern to be specific for melanoma, whereas the parallel furrow, latticelike, and fibrillar patterns are characteristic of benign acral nevi.[8]

In this systematic review and meta-analysis, we synthesized findings from 41 studies encompassing more than 9600 acral melanocytic lesions. Our objective was to identify the dermoscopic features that most reliably distinguish ALMs from acral nevi and expand clinical characterization of ALM by comparing these diagnostic patterns across diverse populations, including patients with skin of color.

Key Points

Question Which dermoscopic features most reliably distinguish acral lentiginous melanoma from benign acral nevi?

Findings In this systematic review and meta-analysis of 41 studies that included 8845 nevi and 801 melanomas, the parallel ridge and multicomponent features were statistically associated with acral lentiginous melanoma, whereas the parallel furrow and latticelike features were significantly associated with benign acral lesions.

Meaning Beyond the previously established ridge and furrow criteria, this systematic review and meta-analysis demonstrates the diagnostic relevance of multicomponent and latticelike features, supporting earlier detection and the standardization of dermoscopic evaluation of acral lesions.

Methods

This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[9] The review protocol was registered with Open Science Framework.

Study Identification

A medical research librarian (D.P.F.) developed and executed comprehensive search strategies across MEDLINE (Ovid), Web of Science Core Collection (Clarivate), the Cochrane Library (Wiley), and PubMed for studies published between January 1, 1970, and March 31, 2023. Controlled vocabulary (medical subject headings [MeSH] and Emtree) and natural language terms were used to capture concepts associated with acral melanoma, nevi, and diagnostic techniques. Searches were limited to English-language and human studies. A validated animal study filter was applied to Embase to exclude nonhuman studies. Search sensitivity was verified by ensuring retrieval of known and relevant citations. Records were deduplicated in EndNote X9 (Clarivate) and screened using Rayyan. The complete search strategy can be found in eTable 1 in Supplement 1.

Study Selection

Inclusion and exclusion criteria are detailed in eTable 2 in Supplement 2. Eligible studies reported patients with acral lesions that were identified as acral melanoma or acral nevi based on clinical or dermoscopic criteria. The primary outcomes of interest were the prevalence of the following dermoscopic patterns: parallel furrow, latticelike, fibrillar, homogeneous, globular, reticular, transition (lesions exhibiting features of more than 1 pattern within the same lesion), nontypical (lesions not conforming to established patterns), parallel ridge, multicomponent (lesions showing multiple distinct dermoscopic structures simultaneously), and other (rare or inconsistently defined dermoscopic features). No limits were applied regarding data sources or outcome measures. Only studies that explicitly stated the number (or proportion) of lesions that corresponded to each dermoscopic pattern were included. Records without this level of detail were excluded.

Data Extraction

Two reviewers (C.O. and J.D.) independently screened records by title and abstract, with disagreements resolved through consensus or adjudication (H.Q.D.). Five reviewers (C.O., T.A.B., A.A., A.T., and N.G.) independently assessed the full texts of studies that passed initial screening, with discrepancies resolved by consensus or adjudication (H.Q.D.). Eight reviewers (C.O., T.A.B., A.A., A.T., N.G., P.L.H., H.S., and H.Q.D.) independently extracted study characteristics, patient demographic characteristics, lesion features, dermoscopic findings, and clinical outcomes using a standardized template.

Quality Assessment

Risk of bias was assessed according to study design using the following validated instruments: the Newcastle Ottawa Quality Assessment Scale (NOS) for cohort studies, the QUADAS-2 tool for diagnostic accuracy articles, and the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for cross-sectional

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studies.[10-12] Publication bias was evaluated using Deeks funnel plots, in which asymmetry suggests overrepresentation of small studies that report larger effects.[13] In the absence of bias, smaller studies scatter widely at the bottom of the plot, with larger studies clustering at the top. Five investigators (T.A.B., C.O., J.D., P.L.H., and H.Q.D.) performed all quality assessments independently, with discrepancies resolved by discussion and consensus. Only studies that met acceptable thresholds of methodological rigor based on these tools were included in the final synthesis.

Cohort Studies (NOS)

The NOS assessed 3 main domains: selection, comparability, and exposure/outcome assessment (maximum score of 9). For selection to be deemed representative, included case patients had to be comparable with the average population and selected through reliable methods (eg, medical records). Comparability required appropriate matching and adjustment for potential confounders, particularly age and sex. Exposure had to be ascertained through validated approaches with sufficient follow-up. Studies were downgraded if outcomes were present at baseline or if confounding was not addressed.

Diagnostic Accuracy (QUADAS-2)

For diagnostic accuracy studies, the QUADAS-2 tool was used to evaluate risk of bias and concerns regarding applicability across 4 domains: patient selection, index test (dermoscopy), reference standards (histopathologic diagnosis), and flow/timing (interval and consistency between dermoscopic and histopathologic assessment). Each domain was rated as low, high, or unclear risk. Studies received an overall judgement for risk of bias (low or at risk) and concern regarding applicability (low concern or concern regarding applicability). A study was designated as low risk if all domains were rated low. Any study judged as high or unclear in 1 or more domains was categorized as having a potential risk of bias or concern regarding applicability.

Cross-Sectional (JBI Checklist)

Cross-sectional studies were critically appraised using the JBI Checklist, comprising 8 items associated with methodological rigor. Elements included clarity of inclusion criteria, use of standardized dermoscopic methods, objective and reliable outcome measurement, identification and control of cofounding variables, and use of appropriate statistical methods. Each item was rated as yes, no, or unclear. Studies were considered high quality if they met most criteria and demonstrated valid outcome measurement and appropriate statistical analysis.

Statistical Analyses

All statistical analyses were performed in R (version 4.3.1; R Foundation) by a research statistician (R.L.B.). A 5% significance level was used for all statistical tests, with no adjustments for multiple testing. For each dermoscopic feature, heterogeneity across studies was assessed with Cochran Q and Higgins I^2 statistic.[14] The Cochran Q serves as a significance test of the null hypothesis that all studies share the same effect.[15] Forest plots were generated to visualize the prevalence of features and pooled proportions using a random-effects model.[16] Comparisons between melanoma and nevi groups were made with a random-effects regression model. Publication bias was evaluated via Deek funnel plots, in which asymmetry suggests overrepresentation of small studies that report larger effects.[13] In the absence of bias, smaller studies scatter widely at the bottom of the plot, with larger studies clustering at the top.

Results

Literature Search and Included Studies

The literature search identified 5925 unique records for screening, of which 255 (0.4%) underwent full-text review. Of these, 41 articles (16%) met inclusion and exclusion criteria (**Table 1**[17-57]). Across these 41 studies, 8845 acral nevi (from 35 studies) and 801 acral melanomas (from 18 studies) were evaluated for unique dermoscopic features. Of the included benign acral nevi, 2182 (25%) were confirmed with histology and 702 (8%) were congenital melanocytic nevi. Forty-five melanomas were described as amelanotic or hypomelanotic across the included records. Vascular dermoscopic structures were reported in 74 lesions overall, with 29 (39.2%) corresponding to amelanotic or hypomelanotic ALMs. The full screening process and exclusion criteria are summarized in the PRISMA diagram (eFigure 1 in Supplement 1), with detailed explanations of excluded studies provided in the eResults in Supplement 1.

Quality Assessment

Funnel plots demonstrated minimal evidence of publication bias in melanoma studies across most dermoscopic features, although some asymmetry was noted for fibrillar and multicomponent patterns. In contrast, nevi studies frequently exhibited asymmetry, suggesting potential bias, particularly for parallel furrow, latticelike, and nontypical patterns. Quality assessment data are summarized in eTables 3 to 5 in Supplement 1.

Cohort Studies (NOS)

Among the 5 cohort studies, 2 achieved an NOS score of 7,[19,45] indicating good quality and low risk of bias, while 3 were scored a 6,[25,32,48] reflecting moderate quality and low to moderate risk of bias. Most studies used representative cohorts and reliable exposure measurements and selected controls appropriately. Common limitations included retrospective design, limited follow-up, and inadequate adjustment for cofounders.

Diagnostic Accuracy Studies (QUADAS-2)

Of the 9 diagnostic accuracy studies, most showed low risk of bias for the index test (dermoscopy) and reference standard (histopathology). The primary sources of bias were associated with patient selection, particularly nonrandom sampling, and lack of histopathologic confirmation for the control lesions. Applicability concerns most often arose from variability in dermoscopy equipment and lack of masking between the index test and reference standard.

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Table 1. Characteristics of Studies Included[a]

Source	Study type	Country/region	Lesions, No.	
			Benign	Malignant
Ahmadabad et al,[17] 2011	Cross-sectional	Iran	47	12
Altamura et al,[18] 2007	Cross-sectional	Italy	230	0
Altmura et al,[19] 2006	Cohort	Italy	723	10
Barquet et al,[20] 2013	Cross-sectional	Uruguay	158	0
Elwan et al,[21] 2016	Cross-sectional	Egypt	223	1
Emiroglu et al,[22] 2017	Cross-sectional	Turkey	210	0
Esener et al,[23] 2019	Cross-sectional	Turkey	143	0
Fagnoli et al,[24] 2014	Cross-sectional	Italy	75	0
Frances et al,[25] 2015	Cohort	Spain	9	0
Gill et al,[26] 2015	Cross-sectional	US	43	0
Gonzales-Ramirez et al,[27] 2018	Cross-sectional	Mexico	1004	0
Han et al,[28] 2020	Cross-sectional	South Korea	0	21
Ishihara et al,[29] 2006	Cross-sectional	Japan	2	20
Iyatomi et al,[30] 2008	Cross-sectional	Japan, Italy, and Austria	169	30
Karaca et al,[31] 2021	Cross-sectional	Turkey	206	0
Koga et al,[32] 2011	Cohort	Japan	0	17
Kokgil et al,[33] 2012	Cross-sectional	Turkey	419	0
Lallas et al,[34] 2015	Diagnostic accuracy	Austria, France, Greece, Japan, and Italy	472	131
Lallas et al,[35] 2014	Cross-sectional	Austria, France, Japan, and Italy	0	118
Li et al,[36] 2019	Cross-sectional	China	217	0
Madankumar et al,[37] 2016	Cross-sectional	US	626	0
Malvehy et al,[38] 2004	Cross-sectional	Spain	210	0
Manci et al,[39] 2022	Cross-sectional	US	0	8
Mikoshiba et al,[40] 2019	Cross-sectional	Japan	331	0
Minagawa et al,[41] 2011	Cross-sectional	Japan	24	0
Minagawa et al,[42] 2013	Cross-sectional	Japan	56	0
Minagawa et al,[43] 2013	Cross-sectional	Japan	420	0
Miyazaki et al,[44] 2005	Cross-sectional	Japan	298	0
Mun et al,[45] 2018	Cohort	South Korea	0	75
Nagashima et al,[46] 2011	Cross-sectional	Japan	69	0
Oguchi et al,[47] 1998	Diagnostic accuracy	Japan	108	22
Ozdemir et al,[48] 2007	Cohort	Turkey	188	0
Ozdemir et al,[49] 2018	Diagnostic accuracy	Turkey	0	46
Phan et al,[50] 2010	Diagnostic accuracy	France	0	66
Saida et al,[51] 1995	Diagnostic accuracy	Japan	77	0
Saida et al,[52] 1997	Diagnostic accuracy	Japan	72	22
Saida et al,[53] 2000	Diagnostic accuracy	Japan	112	39
Saida et al,[54] 2002	Diagnostic accuracy	Japan	198	60
Saida et al,[55] 2004	Diagnostic accuracy	Japan	609	103
Savas et al,[56] 2020	Cross-sectional	China	474	0
Wawrzynkiewicz et al,[57] 2019	Cross-sectional	Poland	624	0
Total	NA	NA	8845	801

Abbreviation: NA, not applicable.

[a] Summary of study characteristics, including first author, year, study type, country or region, and number of benign (acral nevi) and malignant (acral lentiginous melanoma) lesions. A total of 41 studies encompassing 8845 benign and 801 malignant lesions were included in the meta-analysis.

Cross-Sectional (JBI Checklist)

Cross-sectional studies comprised most included records (27 of 41 [66%]). Overall, the methodological quality was high. Most studies used standardized dermoscopic protocols, objective interpretation criteria, and valid outcome measures. Statistical analyses were generally appropriate when applicable. Limitations included insufficient reporting of patient selection methods (setting and inclusion/exclusion criteria) and inadequate identification or control of potential cofounders.

Meta-Analyzed Results

The sensitivities, specificities, and odds ratios of dermoscopic features and patterns in distinguishing acral melanoma from acral nevi were evaluated through meta-analysis. Features assessed included parallel furrow, latticelike, fibrillar, homogenous, globular, reticular, nontypical, parallel ridge, multicomponent, and other patterns. A complete summary of these results is presented in **Table 2**.

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Table 2. Pooled Prevalence of Dermoscopic Patterns in Acral Nevi and Acral Lentiginous Melanoma[a]

Pattern	Proportion, No. (%) [95% CI]		Pooled difference, % (95% CI)	*P* value
	Melanoma	Nevi		
Parallel furrow	63 (8.9) [5.3 to 12.5]	4310 (51.8) [46.7 to 56.8]	41.3 (33.1 to 49.5)	<.001
Latticelike	20 (2.7) [1.0 to 4.5]	1019 (13.0) [10.6 to 15.5]	8.9 (4.2 to 13.6)	<.001
Fibrillar	137 (19.0) [12.5 to 25.5]	1188 (14.9) [11.3 to 18.4]	-3.5 (-10.3 to 3.3)	.31
Homogenous	2 (1.1) [-0.7 to 3.0]	469 (8.5) [5.8 to 11.2]	6.0 (-2.0 to 13.9)	.14
Globular	13 (6.2) [-1.6 to 13.9]	325 (5.8) [4.0 to 7.5]	0.3 (-5.6 to 6.2)	.91
Reticular	4 (2.0) [-1.0 to 5.1]	216 (3.6) [1.8 to 5.5]	0.8 (-4.4 to 6.1)	.75
Nontypical	7 (2.4) [0.1 to 4.6]	474 (7.9) [5.5 to 10.4]	3.8 (-2.8 to 10.4)	.26
Parallel ridge	534 (79.6) [71.4 to 87.8]	53 (0.6) [0.3 to 0.9]	-77.3 (-85.0 to 69.6)	<.001
Multicomponent	96 (45.8) [19.3 to 72.2]	64 (5.0) [3.7 to 6.2]	-38.5 (-65.7 to 11.2)	.01
Other	17 (14.4) [8.1 to 20.7]	96 (7.6) [2.0 to 13.3]	-6.8 (-25.1 to 11.6)	.47

[a] Meta-analysis results comparing the prevalence of dermoscopic patterns between acral lentiginous melanoma and benign acral nevi. Data include pooled proportions with 95% CIs, pooled differences, and *P* values.

adBenign-Associated Patterns

The parallel furrow pattern was the most common feature of acral nevi, with a pooled prevalence of 51.8% (95% CI, 46.7%-56.8%) compared with 8.9% (95% CI, 5.3%-12.5%) in melanoma (pooled difference, 41.3%; 95% CI, 33.1%-49.5%; $P < .001$). The latticelike pattern was also significantly more frequent in nevi (13.0%; 95% CI, 10.6%-15.5%) than melanoma (2.7%; 95% CI, 1.0%-4.5%; pooled difference, 8.9%; 95% CI, 4.2%-13.6%; $P < .001$). Therefore, both patterns demonstrated strong specificity for benign acral lesions, with parallel furrow pattern showing 51.8% sensitivity and 91.1% specificity and the latticelike pattern showing 13.0% sensitivity and 97.3% specificity (**Figure 1**).

Melanoma-Associated Patterns

The parallel ridge pattern demonstrated the strongest diagnostic performance for melanoma, with a prevalence of 79.6% (95% CI, 71.4%-87.8%) in ALM compared with 0.6% (95% CI, 0.3%-0.9%) in nevi (pooled difference, -77.3%; 95% CI, -85.0% to -69.6%; $P < .001$), corresponding to 79.6% sensitivity and 99.4% specificity for melanoma (**Figure 2**). Multicomponent structures were also significantly associated with melanoma (45.8%; 95% CI, 19.3%-72.2%) compared with nevi (5.0%; 95% CI, 3.7%-6.2%; pooled difference, -38.5%; 95% CI, -65.7% to -11.2%; $P = .01$), indicating 45.8% sensitivity and 95.0% specificity (**Figure 3**).

Other features, including fibrillar, homogeneous, globular, reticular, and nontypical patterns, showed mixed or nonsignificant associations. For example, fibrillar features were more common in melanoma (19.0%) than nevi (14.9%) but without statistical significance. Homogeneous, globular, reticular, and nontypical features were more often reported in nevi than melanoma (8.5%, 3.6%, and 7.9% vs 1.1%, 2.0%, and 2.4%, respectively), but the differences were not statistically significant. Substantial heterogeneity (I^2 >75%) was observed for parallel furrow, fibrillar, and multicomponent patterns (Deek funnel plots are included in eFigures 2-4 in Supplement 1), while other features demonstrated lower variability across studies.

Demographic Results

Of the 41 included records, 11 (26.8%) reported Fitzpatrick skin type, while 33 (80.5%) reported some form of racial or ethnic demographic information. Eight studies (19.5%) provided no demographic data at all. Only 3 studies analyzed dermoscopic findings specifically in darker skin (Fitzpatrick IV-VI). All studies and their demographic information are demonstrated in eTable 6 in Supplement 1.[26,37,39]

Discussion

While prior research has investigated dermoscopic features that increase clinical suspicion for ALM, most studies have been limited to single institutions and have not reported on larger cohorts or registries.[58] In this systematic review and meta-analysis, we synthesized data from 41 studies encompassing more than 9600 acral lesions across 16 countries, providing one of the most comprehensive evaluations of dermoscopic features of ALM to date. The meta-analytic approach, regarded as the most rigorous method for evidence synthesis, enhanced the reliability of our prevalence estimates and diagnostic comparisons. To further strengthen confidence in the findings, 3 validated quality assessment tools were applied to evaluate risk of bias in each included study. By pooling data from diverse populations and clinical settings, this study generated summative conclusions with greater external validity and established diagnostic benchmarks that extend beyond the limitations of single-center reports.

The parallel ridge pattern emerged as the most sensitive and specific dermoscopic marker for ALM, reinforcing prior reports that established this pattern as the most reliable dermoscopic criterion for melanoma on acral surfaces.[28] However, approximately 20% of ALMs did not exhibit this feature, underscoring that its absence should not be used to exclude cancer.[28] Multicomponent structures were also more frequently associated with melanoma than nevi, representing a moderately sensitive but highly specific feature. Conversely, the parallel furrow and latticelike patterns were most often associated with acral nevi, validating their role as reassuring features when present.[32,55] Eight percent of benign nevi in this review were congenital melanocytic nevi, which generally followed dermoscopic trends of acquired melanocytic nevi but exhibited the crista-dot (2 “peas-in-a-pod”) feature.[41-43] Other patterns, including fibrillar, homogeneous, globular, reticular, and nontypical structures, showed mixed or

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Figure 1. Forest Plot of Dermoscopic Features Associated With Benign Acral Nevi

Composite figure showing the pooled prevalence of dermoscopic patterns significantly associated with benign acral nevi. Error bars represent 95% CIs. Axis scales vary to improve readability and display of data distribution. Representative dermoscopic patterns are observed in acral nevi and melanoma.

[a]Observer 1 for Gonzales-Ramirez et al.[27]

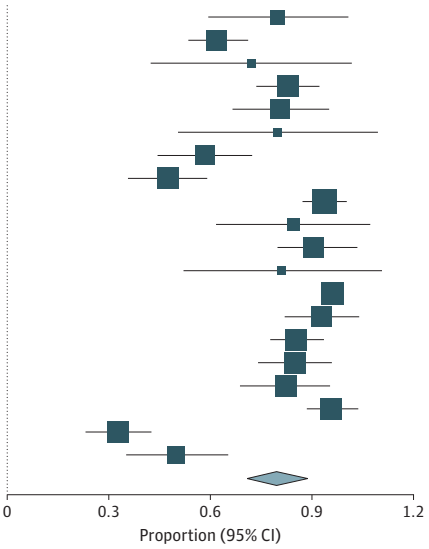
[b]Observer 2 for Gonzales-Ramirez et al.[27]

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Figure 2. Forest Plot of the Parallel Ridge Pattern in Acral Lentiginous Melanoma and Benign Acral Nevi

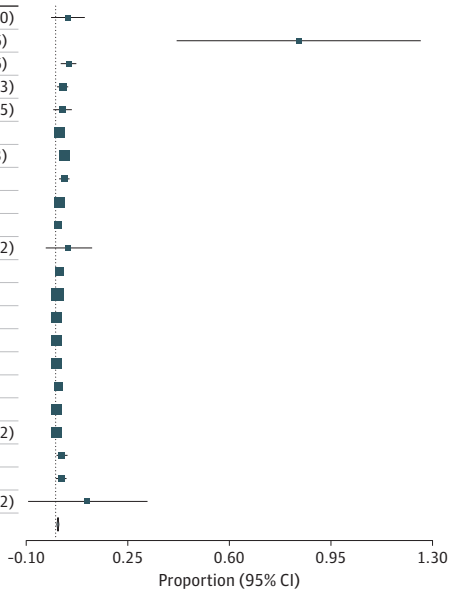
A Parallel ridge pattern in melanoma

Source	Proportion (95% CI)
Ahmadabad et al,[17] 2011	0.83 (0.62 to 1.04)
Lallas et al,[35] (MIS specific) 2014	0.65 (0.57 to 0.74)
Manci et al,[39] 2022	0.75 (0.45 to 1.05)
Mun et al,[45] (melanoma) 2018	0.86 (0.76 to 0.96)
Mun et al,[45] (MIS specific) 2018	0.84 (0.70 to 0.98)
Oguchi et al,[47] (melanoma) 1998	0.83 (0.54 to 1.13)
Ozdemir et al,[48] (MIS specific) 2018	0.61 (0.47 to 0.75)
Phan et al,[50] 2010	0.50 (0.38 to 0.62)
Saida et al,[53] (melanoma) 2000	0.97 (0.91 to 1.03)
Saida et al,[53] (MIS specific) 2000	0.88 (0.65 to 1.10)
Saida et al,[52] (melanoma) 1997	0.94 (0.82 to 1.06)
Saida et al,[52] (MIS specific) 1997	0.83 (0.54 to 1.13)
Saida et al,[54] (melanoma) 2002	0.98 (0.93 to 1.02)
Saida et al,[54] (MIS specific) 2002	0.94 (0.83 to 1.05)
Saida et al,[55] (melanoma) 2004	0.87 (0.78 to 0.95)
Saida et al,[55] (MIS specific) 2004	0.86 (0.75 to 0.97)
Iyatomi et al,[30] 2008	0.83 (0.70 to 0.97)
Koga and Saida et al,[32] 2011	0.97 (0.90 to 1.05)
Lallas et al,[34] 2015	0.33 (0.23 to 0.42)
Lallas et al,[34] (melanoma) 2015	0.50 (0.35 to 0.65)
Random-effects model	0.80 (0.71 to 0.88)



B Parallel ridge pattern in nevi

Source	Proportion (95% CI)
Ahmadabad et al,[17] 2011	0.04 (-0.02 to 0.10)
Ishihara et al,[29] 2006	0.83 (0.41 to 1.26)
Iyatomi et al,[30] 2008	0.04 (0.01 to 0.06)
Karaca Ural et al,[31] (women)2021	0.01 (-0.01 to 0.03)
Karaca Ural et al,[31] (men)2021	0.02 (-0.02 to 0.05)
Kokgil et al,[33] 2021	0 (0 to 0.01)
Lallas et al,[34] 2015	0.02 (0.01 to 0.03)
Li et al,[36] 2019	0.02 (0 to 0.04)
Madankumar et al,[37] (palms) 2016	0 (0 to 0.01)
Madankumar et al,[37] (soles) 2016	0 (0 to 0.01)
Minagawa et al,[41] 2011	0.04 (-0.04 to 0.12)
Minagawa et al,[41] 2013[b]	0.01 (0 to 0.02)
Altamura et al,[19] 2006	0 (0 to 0)
Oguchi et al,[47] 1998	0 (-0.01 to 0.02)
Saida et al,[53] 2000	0 (-0.01 to 0.02)
Saida et al,[54] 2002	0 (0 to 0.01)
Saida et al,[54] 2004	0.01 (0 to 0.02)
Savas Erdogan et al,[56] 2020	0 (0 to 0.01)
Wawrzynkiewicz et al,[57] 2019	0.01 (-0.01 to 0.02)
Elwan et al,[21] 2016	0.02 (0 to 0.04)
Emiroglu et al,[22] 2017	0.02 (0 to 0.04)
Frances et al,[25] 2015	0.11 (-0.09 to 0.32)
Random-effects model	0.01 (0 to 0.01)



Pooled prevalence of the parallel ridge pattern in acral lentiginous melanoma compared with benign acral nevi. Horizontal lines indicate 95% CIs; the diamond represents the pooled estimate. MIS indicates melanoma in situ.

nonsignificant associations, limiting their diagnostic utility in isolation.

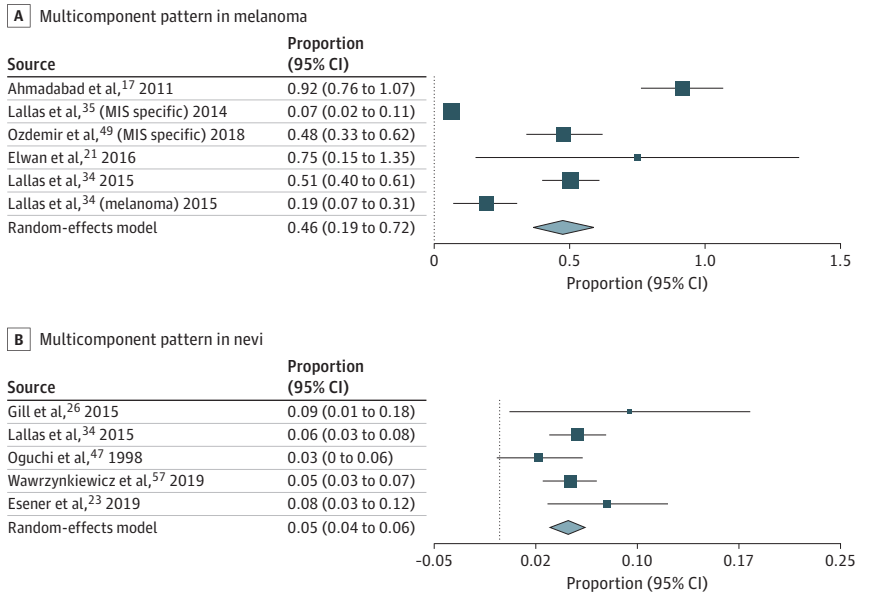
Recent investigations have identified the parallel ridge pattern in benign acral nevi, specifically in patients with darker phototypes.[59] A small cohort of 8 Black patients with acral melanoma demonstrated the parallel ridge pattern in all cases; however, it was also observed in adjacent normal skin in 7 of the 8 patients. Although the normal skin was not biopsy confirmed as benign and the cohort was small, these findings raise the possibility that the parallel ridge pattern may occasionally appear in benign acral skin in patients with darker phototypes and warrants further investigation.[39] Outside of this isolated report, the parallel ridge pattern in benign lesions remained rare (<5%) across all other included records, with no evidence of increased prevalence among patients with darker phototypes.

Clinical Implications

These findings reaffirm the diagnostic importance of the parallel ridge pattern for acral melanoma and the parallel furrow and latticelike patterns for benign nevi, as outlined in the expanded 3-step algorithm. This study further extends the

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Figure 3. Forest Plot of the Multicomponent Dermoscopic Pattern in Acral Lentiginous Melanoma and Benign Acral Nevi



Multicomponent dermoscopic patterns differentiating acral lentiginous melanoma from benign nevi. Error bars indicate 95% CIs. Axis scales vary to improve the readability and display of data distribution. MIS, melanoma in situ.

algorithm by establishing multicomponent structures as an additional valuable feature for identifying ALM. The higher prevalence of multicomponent dermoscopic features in melanoma, coupled with its low frequency in nevi, underscores its clinical utility, despite moderate sensitivity. In contrast, this study found fibrillar patterns to occur with similar frequency in ALM and benign nevi, challenging the algorithm's classification of this feature as reassuring and underscoring that fibrillar structures should not be used in isolation to rule out cancer.

Accordingly, lesions demonstrating parallel ridge and multicomponent features should be prioritized for biopsy specimens, whereas lesions displaying parallel furrow and latticelike features may be more appropriate candidates for clinical monitoring unless additional concerning features are present. Because these benign patterns were still observed in few melanomas, clinical context and evolution should always guide biopsy decisions. Beyond their immediate diagnostic use, the pooled estimates from this review provide evidence-based benchmarks that can be incorporated into dermoscopy training, establishing clear reference points for residency curriculum, dermatology education, and general practice.[60,61] Last, the diagnostic metrics reported in this article may also serve as reference standards for evaluating emerging diagnostic technologies, such as artificial intelligence–based image analysis, ensuring that such tools meet clinically relevant thresholds before integration into practice.[62]

Limitations

This study had several limitations. First, heterogeneity in diagnostic performance was observed across studies, likely reflecting differences in study design, examiner expertise, and reporting standards. In addition, definitions of dermoscopic structures were not uniform, complicating direct comparisons. Many included studies were retrospective, single-institution analyses with relatively small cohorts, limiting generalizability. Additionally, skin type demographic characteristics were frequently underreported. More than 80% of included studies reported race or ethnicity, but only 27% specified the Fitzpatrick phototype, which can vary widely within racial and ethnic groups. This heterogeneity limited the ability to evaluate population-specific dermoscopic patterns. Finally, as with all literature reviews, our analysis was limited by the quality and availability of the published literature.

Future research should adopt standardized dermoscopic terminology and reporting practices, ideally in prospective, multicenter studies that rigorously document patient demographic characteristics and skin type. Future studies should also assess additional clinical and dermoscopic clues in atypical cases, such as melanomas with a parallel furrow pattern or nevi with a parallel ridge pattern, to refine biopsy decision-making. Emerging evidence suggests that the sensitivity of certain dermoscopic features may vary across populations, underscoring the need to clarify these differences to guide clinical practice.[59] As artificial intelligence–based diagnostic tools continue to evolve, ensuring diverse and representative training datasets will be essential to improve diagnostic accuracy while minimizing the risk of patient harm.

Conclusions

This systematic review and meta-analysis confirms that a hierarchy of dermoscopic features exists in the diagnosis of ALM. The parallel ridge and multicomponent patterns demonstrated the most significant diagnostic value for identifying ALM, whereas parallel furrow and latticelike patterns were most commonly associated with benign acral nevi.

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Although ALM represents only 2% to 3% of melanoma cases, it often carries a worse prognosis, partially due to delayed recognition and the heterogeneity of its clinical presentation.[58] By clarifying which dermoscopic patterns most reliably distinguish ALM from benign acral lesions, this study provides evidence-based criteria to support earlier recognition, facilitate timely treatment, and improve clinical outcomes.

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